
Image Denoising: How the Field Actually Evolved

Not ten papers in order. One technical story: what each method could not do, what the next one changed, and which line of code changed to do it.

THE QUESTION

Why did each innovation have to be invented?

ANCHOR PAPER

DnCNN — Zhang et al. IEEE TIP 26(7), 2017

PRESENTERS

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REFERENCE

100-slide master deck, submitted separately

Four questions we will ask about every method

Each algorithm gets the same four-part treatment, so the comparison between them is structural rather than anecdotal. The methods that were state of the art get a fifth.

1 What could the previous method not do?

Every innovation in this field was a response to a named, measured failure — usually named by the paper that fixed it, which is what makes this timeline evidenced rather than imposed.

2 What changed technically?

The algorithmic change: a different prior, a different training target, a different operator. Stated precisely enough that you could implement it.

3 Which line of code changed?

A Before → After from the reference implementations. Contribution → algorithmic change → implementation change → measured improvement, as one chain.

4 What did it buy, and what did it cost?

The measured improvement, the applications it opened, and the new limitation it created — because every step in this story created the next problem.

5 And for the SOTA methods, a fifth: **what made it state of the art, on which benchmark and in which years** — and why it did not make the others obsolete.

One equation, three unknowns, eighteen years

Given only y , infinitely many splits into $x + v$ are arithmetically valid. Every method in this talk supplies the missing information in a different way — that supplied information is called the prior, and the whole story is about where it lives.

$$y = x + v$$

y — what the sensor produced · x — the clean image we want · v — the noise

- 01 The classical answer writes the prior down as a formula: minimise $\frac{1}{2}\|y - x\|^2 + \lambda \cdot \Phi(x)$. You can read Φ , differentiate it, and argue about it.
- 02 The deep answer replaces Φ with a tensor of learned weights. You cannot read it, but you can download it — and it runs in 60 ms.
- 03 The self-supervised answer takes the prior partly from the test image itself, so no clean data is needed at all.
- 04 **Where the prior lives, and who writes it, is the single axis this whole talk runs along.**

What state of the art means in this talk

The phrase is used loosely in most presentations. We attach three qualifiers to it, because that is what makes the SOTA comparison later in the talk meaningful rather than rhetorical.

A SOTA CLAIM NEEDS THREE THINGS

- 1 **A benchmark and a metric.** Best denoiser is meaningless. Highest PSNR on BSD68 at $\sigma=25$ is a claim you can check.
- 2 **A comparison protocol.** Two papers reporting on the same dataset can still disagree, so wherever a row compares two methods we take both numbers from one table.
- 3 **Honesty about what is not reported.** No paper here states how long it held a position, so this talk does not claim SOTA windows — only the publication date and which table each method leads.

WHICH IS WHY THERE IS NO SINGLE WINNER

- Different benchmarks measure different capabilities: **BSD68** measures average-case Gaussian removal, **Urban100** measures long-range self-similarity, **SIDD** measures real sensor noise.
- A method can lead one and trail another. **Restormer**, **NAFNet** and **MambaIR** each lead a different table today.
- We return to this on **slide 37** with the numbers, because why there are several SOTAs is the question this talk is built to answer.

benchmark definitions: **DnCNN17** §I, §III · **Restormer22** §4.1 · **NAFNet22** §4.1

2007

2014

2017

2022

2025

Three numbers, three opinions — why PSNR alone is not enough

The brief asks for PSNR, SSIM and perceptual metrics side by side. This talk reports PSNR because its sources do — but every PSNR claim names its table and its σ . This slide is why that discipline matters.

PSNR

$$\text{PSNR} = 10 \cdot \log_{10}(\text{MAX}^2 / \text{MSE})$$

Pixel-wise fidelity in decibels. Optimal for the conditional mean — which is smoother than any plausible clean image. Two visibly different images can share one PSNR.

SSIM

$$\text{SSIM}(x, y) = ((2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)) / ((\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2))$$

Structure-aware: luminance, contrast, correlation. Catches blur that PSNR forgives; still a low-level metric.

LPIPS

$$d(x, \hat{x}) = \sum_1 \|w_1 \odot (F_1(x) - F_1(\hat{x}))\|^2 / (H_1 \cdot W_1)$$

Distance in calibrated deep-feature space. Matches human judgment; standard in super-resolution, absent from denoising leaderboards.

- 01 Noise2Noise names the mechanism itself: “*L2 training results in spatial blurriness*.” While leaderboards rank by PSNR, the incentive points the wrong way. [N2N18 §2]
- 02 Our rule for the whole talk: every PSNR number is reproduced from a **named table** (DnCNN17 Table II, Restormer22 Tables 4–6 ...) at a stated dataset and σ — never recomputed by us.
- 03 For applications the metric that matters is **downstream**: detection or segmentation IoU on denoised input. Denoising is preprocessing, not the product — see slide 40.

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PART 1 / 5

| BASELINE • 2007 - 2014

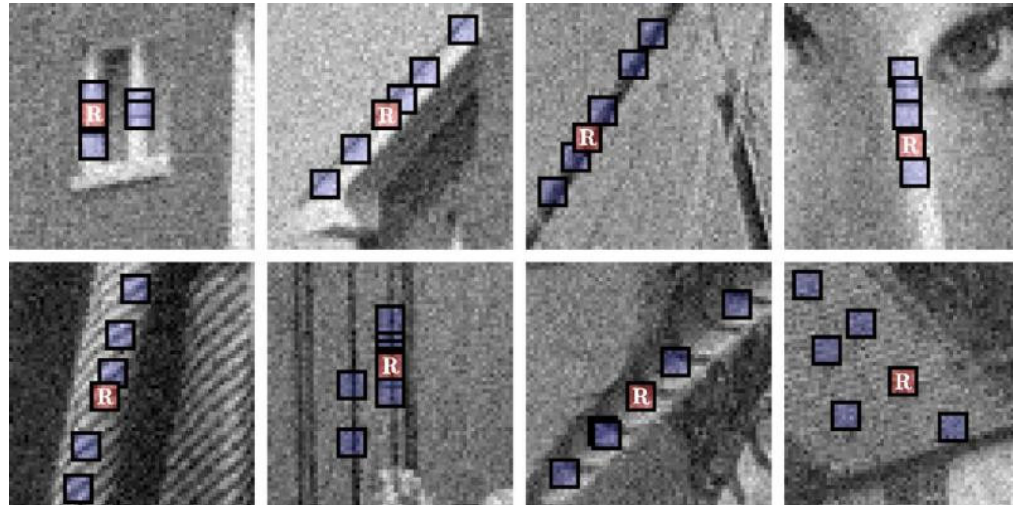
The Hand-Written Prior

- 01 BM3D — the baseline everything is measured against
- 02 WNNM — the classical ceiling
- 03 Bottleneck 1 — why the era ended

Where the field started: **a formula a human wrote, re-solved from scratch for every image.**

The innovation: filter patches together, not alone

Dabov et al., IEEE TIP 2007. Previous transform-domain methods shrink each patch independently. BM3D groups similar patches first, creating a third axis along which the signal is nearly constant and the noise is not.



Block matching in practice: the reference patch R and the similar patches BM3D groups with it before collaborative 3-D filtering. [BM3D07 · Fig. 2]

WHAT CHANGED

Block-match similar patches into a 3-D stack, transform the stack, hard-threshold, invert, aggregate — then repeat using the cleaner estimate as a **Wiener pilot**.

WHY IT WORKS

The paper own ablation: even a transform that is random apart from the DC loses only 0.1–0.4 dB. **The grouping does the work, not the transform.** [BM3D07 §V]

The innovation: filter patches together, not alone

CONTINUED

IMPACT

BSD68, $\sigma=25$: **28.57 dB** at **2.85 s** per 512×512 image [DnCNN17 Table II, Table IV]. Became the **reference line every paper for a decade had to beat**.

STILL IN USE

Appears as a live baseline in **Noise2Noise (2018)**, **Noise2Void (2019)**, **ZS-N2N (2023)** and **Restormer (2022)** — a seventeen-year-old algorithm is still what new zero-shot methods must beat, **because it needs no training data**.

APPLICATIONS

The zero-shot comparison in those same papers, and the fallback where there is neither training data nor a GPU. On real sensor noise it still beats an AWGN-trained DnCNN by **1.99 dB** — BM3D 25.65 vs DnCNN 23.66 on SIDD, both from Restormer22 Table 6.

28.57 dBBSD68 $\Sigma=25$ **2.85 s**

512×512

+1.99 dB

VS DNCNN ON SIDD

WNNM vs BM3D — the same grouping, a better estimator

Gu et al., CVPR 2014. WNNM keeps BM3D non-local grouping and replaces what happens to the group. The first clean Before → After in the story, entirely inside one step.

$$\hat{X} = \operatorname{argmin}_X \frac{1}{2} \|Y - X\|_F^2 + \sum_i w_i \cdot \sigma_i(X), \quad w_i \propto 1/\sigma_i$$

the weighted nuclear norm: large singular values carry the image and are shrunk less [WNNM14 §3]

ASPECT	BM3D (2007)	WNNM (2014)	WHY IT MATTERS
Core algorithm	Group patches, shrink in a 3-D transform	Group patches, recover a low-rank matrix	Stacked near-copies are low rank by construction — a stronger structural assumption
Key innovation	Collaborative hard-thresholding	Weighted nuclear norm, $w_i \propto 1/\sigma_i$	Standard NNM shrinks all singular values equally
Implementation change	3-D DCT + threshold	SVD per patch group + per-value shrinkage, iterated	One transform call becomes an iterated SVD — the entire cost difference
Performance	28.57 dB • 2.85 s	28.83 dB • 773.2 s	+0.26 dB for a 271× slowdown
Limitation	Fixed hand-set parameters	Non-convex; still per-image optimisation	Neither can reuse anything learned from one image on the next
Applications	General-purpose, real-time-ish	Offline / quality-critical restoration	The cost makes WNNM impractical for interactive or batch use

This row is the whole argument for what comes next. Seven years of hand-designed priors bought 0.26 dB on BSD68 at $\sigma=25$, and the best method became 271× slower than the second best. The era hit a wall in quality and cost simultaneously.

DnCNN17 Table II, Table IV • WNNM14 §3, Theorem 2



Why the era ended — stated by the paper that ended it

DnCNN introduction names the classical era drawbacks in its own words. Quoting the paper that broke the bottleneck, rather than asserting it ourselves, is what makes this an evidenced timeline.

DRAWBACK 1

The prior is paid for once per image

“Prior-based methods involve a complex optimization problem in the testing stage.” Nothing learned on image 1 helps image 2. WNNM: 773 s, every time. [DnCNN17 §I]

DRAWBACK 2

The prior is written by hand

“The models are generally non-convex and involve several manually chosen parameters” — block size, thresholds, λ , different values above and below $\sigma=40$. [DnCNN17 §I]

DRAWBACK 3

Its form limits the ceiling

“Performance is inherently restricted to the specified forms of prior.” You cannot exceed what your formula can express, however well you tune it. [DnCNN17 §I]

DRAWBACK 4

One model per noise level

“Bridge methods train a specific model for a certain noise level and are limited in blind image denoising.” In a real camera, σ is never known. [DnCNN17 §I]

DnCNN17 §I — all four quoted from the paper introduction

2007

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PART 2 / 5

THE BREAK • 2016 - 2017

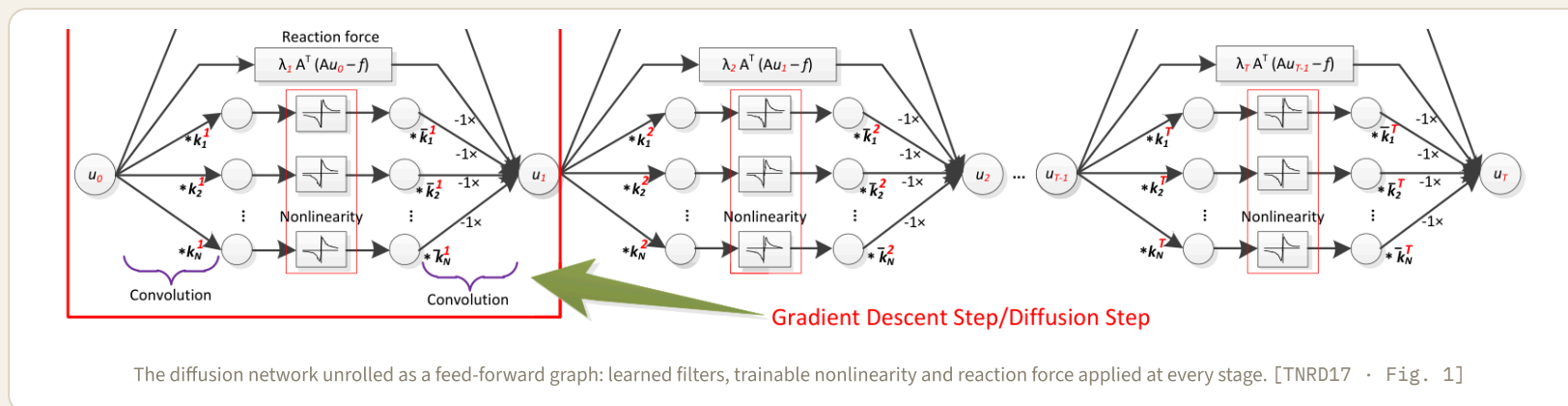
The Prior Becomes a File

- 01 TNRD — the bridge nobody skips
- 02 DnCNN — innovation, implementation, impact
- 03 Bottlenecks 2 and 3

One architecture answers all four drawbacks at once — **and creates the two problems the rest of the talk is about.**

Unrolling — the step that made the transition continuous

Chen & Pock, IEEE TPAMI 2017. Worth ninety seconds because it is the conceptual bridge: it shows that a classical optimiser and a neural network are the same object viewed differently, which is why the move to deep learning was not a rupture.



ASPECT	CLASSICAL OPTIMISER	TNRD (UNROLLED)	WHY IT MATTERS
Iteration count	Run to convergence, unbounded	Truncated to a fixed K (usually < 10)	Runtime becomes a constant — the first half of bottleneck 1 is gone
Parameters	Same hand-set rule every iteration	Every stage has its own learned filters and nonlinearities	The prior is no longer one formula; it is K different learned ones
What is learned	Nothing	Filters and influence functions, jointly, from data	Earlier diffusion models fixed the filters and chose the diffusivity
Implementation	while loop to a tolerance	A fixed stack trained by back-propagation	Structurally a CNN — well-suited for parallel computation on GPUs

Unrolling — the step that made the transition continuous

CONTINUED

ASPECT	WNNM (CLASSICAL CEILING)	TNRD (UNROLLED)	WHY IT MATTERS
Performance	28.83 dB • 773 s	28.92 dB • 0.032 s (GPU)	Best quality-per-second of the era
Limitation	—	Still one model per noise level	Blind denoising remains unsolved

KEEP THIS IN MIND FOR THE NEXT SLIDE

DnCNN derives part of its own justification from a **one-stage TNRD** — the two are directly connected, not merely adjacent in time.

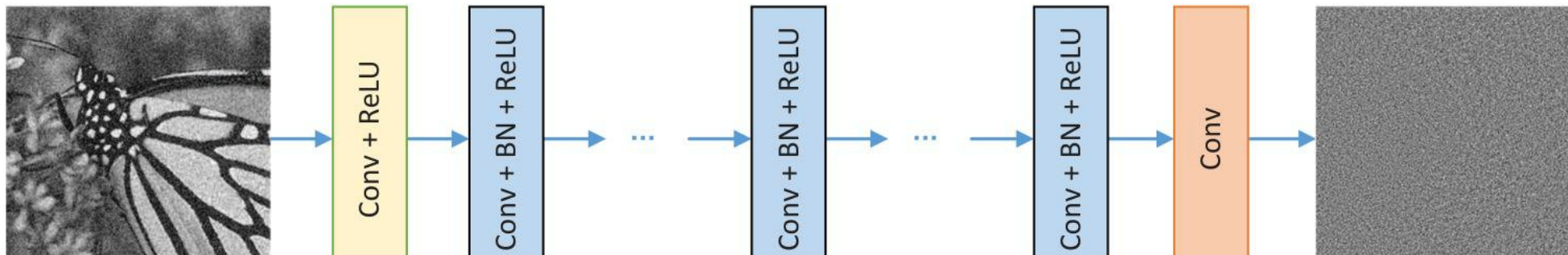
28.92 dB • 0.032 s

quality of WNNM at **1/24,000** of its runtime [DnCNN17 Tables II, IV]



DnCNN vs TNRD — remove the optimiser entirely

Zhang et al., IEEE TIP 26(7), 2017. TNRD unrolled the optimiser; DnCNN throws away the optimisation heritage and keeps only the network. The technical change is small; the consequence is the largest single step in this story.



The DnCNN architecture: Conv(+BN)+ReLU layers predict the residual noise, not the clean image. [DnCNN17 • Fig. 1]

ASPECT	TNRD (2017 TPAMI)	DNCNN (2017 TIP)	WHY IT MATTERS
Core algorithm	K diffusion stages, PDE-derived	Plain deep CNN: Conv-BN-ReLU ×17, no pooling	The architecture no longer has to resemble an optimiser at all
What the net outputs	The restored image	The noise — $x = y - R(y)$	Residual learning: the do-nothing solution moves to zero, which is trivially reachable
Key enabler	Learned influence functions	Residual learning + batch normalisation, together	Neither works nearly as well alone — the paper most-cited empirical finding
Noise levels	One model per σ	DnCNN-B: one blind model, $\sigma \in [0, 55]$	Drawback 4 solved; no noise estimate needed at test time

DnCNN17 §II-A, §III-A, §III-C, §III-D, Table II, Table IV • TNRD17

2007

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2025

DnCNN vs TNRD — remove the optimiser entirely CONTINUED

ASPECT	TNRD (2017)	DNCNN (2017)	WHY IT MATTERS
Performance	28.92 dB • 0.032 s	29.23 dB • 0.060 s (blind: 29.16)	+0.66 / +0.66 / +0.61 dB over BM3D at $\sigma = 25 / 15 / 50$ [DnCNN17 Table II]
Applications	Gaussian denoising	One model for denoising + super-resolution + JPEG deblocking	Redefine v and the same network solves three tasks [DnCNN17 §III-D, Table V]

+0.66 dB

OVER BM3D • BSD68 $\Sigma=25$

$\sigma=15$: +0.66 • $\sigma=50$: +0.61 — one paper took more than seven years of classical refinement (WNNM +0.26, TNRD +0.35, EPLL +0.11) [DnCNN17 Table II]

12,900×

FASTER THAN WNNM • 512×512

773.2 s CPU → 0.060 s GPU. The device differs and the paper says so; the order of magnitude is the point [DnCNN17 Table IV]

3 tasks

ONE MODEL, ONE LOSS

DnCNN-3 handles denoising, SR and JPEG deblocking, beating a specialist baseline on all three [DnCNN17 Table V]

DnCNN17 Table II, Table IV, Table V



The innovation is one line of code

cszn/KAIR, models/network_dncnn.py, by DnCNN own first author. The architecture is a list comprehension; the contribution is the return statement.

KAIR/models/network_dncnn.py — the whole network

```
m_head = B.conv(in_nc, nc, mode=...)
m_body = [B.conv(nc, nc, mode=...)
           for _ in range(nb-2)] # ← the entire network
m_tail = B.conv(nc, out_nc, mode=...)

def forward(self, x):
    n = self.model(x)
    return x-n                # ← the contribution

# BEFORE (discriminative CNNs, ~2012–2016):
#     return self.model(x)      → learn  $F(y) = x$ 
# AFTER (DnCNN, 2017):
#     return x - self.model(x) → learn  $R(y) = v$ 
```

- 01 **Why that one change matters.** From ResNet: a mapping close to the identity is hard to optimise. And y is much more like x than it is like v — so learning the image forces the network toward an identity transform, which seventeen ReLU layers have no natural way to express.
- 02 **Learning the noise** moves the trivial solution to zero, reachable by driving weights toward zero. Same function class, far easier optimisation landscape.
- 03 **The synergy, in one sentence.** Residual learning strips the image out of the hidden layers, making activations Gaussian-like and content-independent — exactly the regime batch-norm batch statistics assume. Without residual learning, BN has a certain adverse effect on convergence. [DnCNN17 §III-C]
- 04 **Where the prior went.** Nowhere in this file. It is in the .pth checkpoint this class will load — the moment the prior stopped being readable and started being downloadable.

measured — cszn/KAIR, verbatim · DnCNN17 §II-A, §III-C

2007

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What made it state of the art, and for how long

A SOTA claim with its qualifiers attached: the benchmark, the margin, and the context.

+0.66 dB

OVER BM3D • BSD68 $\Sigma=25$

$\sigma=15$: +0.66, $\sigma=50$: +0.61. Seven years of classical refinement had bought +0.11–0.39 dB (EPLL, WNNM, TNRD); DnCNN took more in one paper. [DnCNN17 Table II]

12,900×

FASTER THAN WNNM • 512²

773.2 s CPU → 0.060 s GPU. The device differs and the paper says so; the order of magnitude is the point. [DnCNN17 Table IV]

3 tasks

ONE MODEL, ONE LOSS

DnCNN-3 handles denoising, super-resolution and JPEG deblocking by redefining v , beating a specialist baseline on all three. [DnCNN17 §III-D, Table V]

- 01 **And it did it blind.** DnCNN-B reaches 29.16 dB at $\sigma=25$ without being told σ — within 0.07 dB of the noise-specific model. Drawback 4 closed. [DnCNN17 Table II]
- 02 **Applications it opened:** real-time denoising in imaging pipelines; the .pth checkpoint as a shippable artefact; denoising as a reusable plug-and-play prior inside other solvers. [SCUNet23 §1]
- 03 **What it was not.** On real sensor noise DnCNN was never the leader — 23.66 dB on SIDD [Restormer22 Table 6]. That gap is slide 41 subject (Open 1).

DnCNN17 Table II, Table IV, §III-D • SCUNet23 §1 • Restormer22 Table 6

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What DnCNN broke, and the two bills it left

This slide is the hinge of the whole talk. Everything after it is the field paying off these two debts — paid off by two different communities, independently, and never merged.

BOTTLENECK 2 — IT NEEDS CLEAN DATA

The loss is $\|R(y) - (y - x)\|^2$. It requires x .

For fluorescence microscopy, cryo-EM, MRI and astronomy a clean ground truth does not exist and cannot be acquired — you cannot photograph the same cell noiselessly.

So the most valuable applications of denoising were the ones DnCNN could not serve.

→ Answered by Branch A — the data branch. Part 3, slides 21–26.

BOTTLENECK 3 — IT CANNOT SEE FAR ENOUGH

Receptive field is 35×35 pixels, chosen deliberately to match EPLL, the smallest effective patch size in the field [DnCNN17 Table I]. WNNM effective patch is 361×361 .

Visible in DnCNN own Table III: on Barbara it loses to BM3D by 0.71 dB — the one Set12 image dominated by repetitive texture.

The paper explains it: repetitive images meet well with the non-local similarity prior that a 35×35 convolution stack cannot reach.

→ Answered by Branch B — the architecture branch. Part 4, slides 28–34.

A limitation visible in 2017, in the winning paper own results table, that took until 2022 to fix. That is what an honest evolution story looks like.

2018: the field forks, and never merges

Two debts, two communities, two independent answers. Neither branch has adopted the other advance — which is itself one of the findings of this project, and the source of the open problem we end on.

BRANCH A · BOTTLENECK 2 — CHANGE WHAT THE NETWORK IS SHOWN

Noise2Noise → **Noise2Void** → **ZS-N2N** → **P2N**

The architecture stays almost exactly where DnCNN left it; what moves is the training target. Each paper removes one more requirement until nothing is needed but the noisy image itself.

BRANCH B · BOTTLENECK 3 — CHANGE WHAT THE NETWORK IS

Restormer, **NAFNet**, **SCUNet**, **MambaIR**

Supervision stays exactly as DnCNN had it — paired data, same loss family. What moves is the operator that replaces the 3×3 convolution.

THE EVIDENCE THEY NEVER MERGED — P2N SHIPS THE 2018 NETWORK

In the official P2N repository the class is literally named UNet_n2n_un, with the 2018 Noise2Noise channel widths. Seven years, and the backbone is unchanged. [repository: Li-Tong-621/P2N-plus, measured]

THE OPEN QUESTION — NOBODY HAS COMBINED THEM

As far as we found, no published work trains a Transformer or state-space backbone under self-supervised RDC/DCS supervision. We return to this as a concrete proposal on slide 42.

the structural claim of this project · repository: Li-Tong-621/P2N-plus [measured]

2007

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| BRANCH A • 2018 - 2025

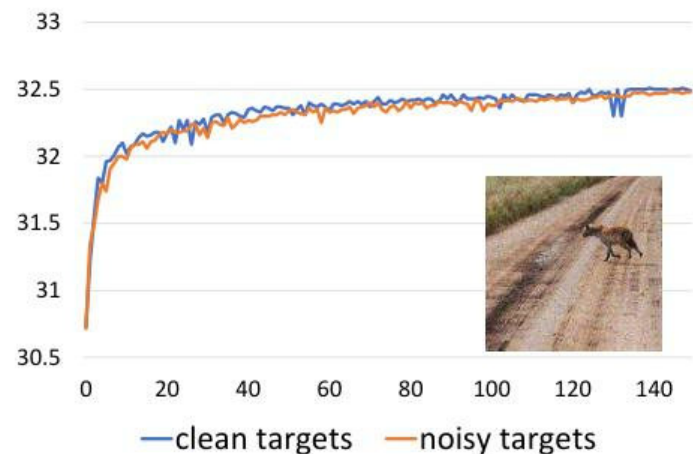
Removing the Clean Image

- 01 Noise2Noise — the statistical result
- 02 Noise2Void — the blind spot
- 03 ZS-N2N and P2N — down to one image
- 04 What self-supervision costs

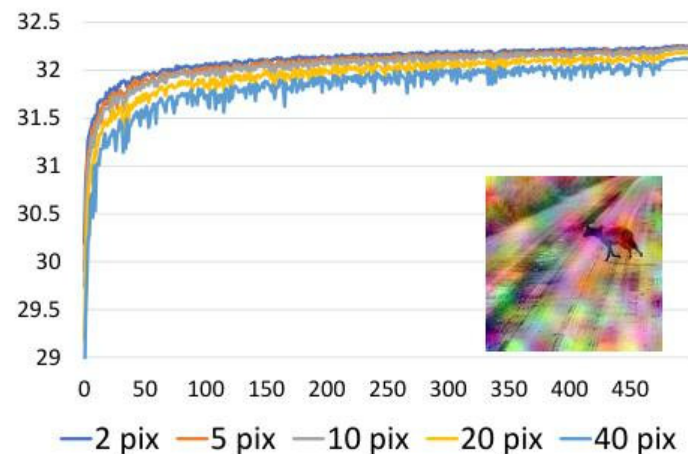
Four papers, each removing exactly one requirement its predecessor could not drop.

A theorem worth one conditional

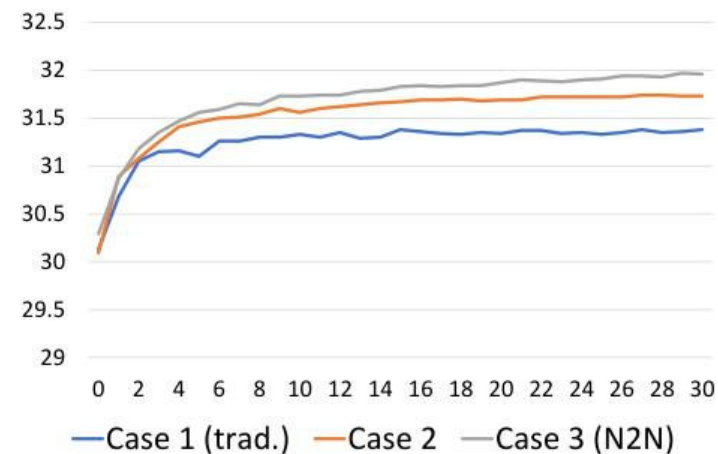
Lehtinen et al., ICML 2018. The most striking ratio in this story: a substantial statistical result whose implementation change is a single branch on which tensor becomes the target.



(a) White Gaussian, $\sigma = 25$



(b) Brown Gaussian, $\sigma = 25$



(c) Capture budget study (see text)

Noise2Noise own result: training on noisy targets (orange) tracks clean-target training (blue) almost exactly. [N2N18 • Fig. 1]

Given a noisy input and a corrupted target whose expectation equals the clean target, L2 regression converges to the same optimum — the corrupted target is a full substitute for ground truth. [N2N18 §2]

A theorem worth one conditional

CONTINUED

THE CHANGE, IN ESSENCE

```
# BEFORE — supervised (DnCNN, 2017)
target = clean          # requires ground truth
loss   = mse(net(noisy), target)

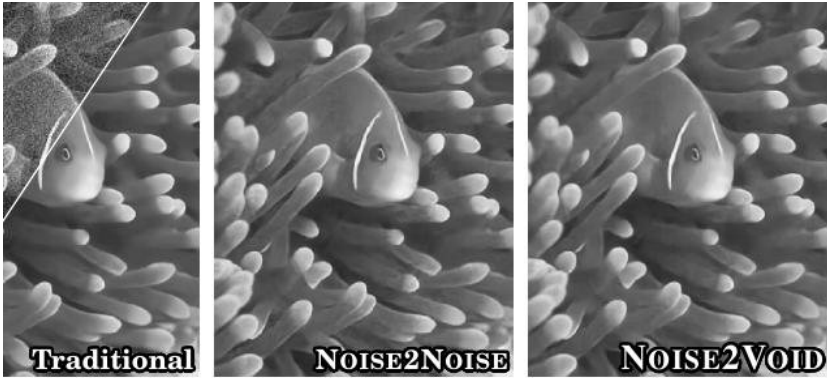
# AFTER — Noise2Noise (2018)
target = second_noisy_shot # ← the entire change
loss   = mse(net(noisy), target)

# network, optimiser, schedule, loss family: all unchanged
```

- 01 **Why it works.** L2 regression learns the conditional mean of the target. The paper states the consequence as a triviality: on expectation, the estimate remains unchanged if we replace the targets with random numbers whose expectations match the targets.
- 02 **So a second noisy photograph** of the same scene is a valid target, because the noise is zero-mean. Given infinite data the solution is identical to clean-target training; for finite data the extra variance shrinks with dataset size.
- 03 **What it does NOT need:** no likelihood model of the corruption, no density model of clean images.
- 04 **Measured cost: 0.02 dB.** Kodak/BSD300/Set14 at $\sigma=25$ — clean targets 31.63, noisy targets 31.61. The clean-data requirement was never actually necessary. [N2N18 Table 1]

N2V vs N2N — removing the second photograph

Krull et al., CVPR 2019. Noise2Noise still needs two independent shots of one scene, which rules out anything that moves and every archival image. N2V removes that too — and pays for it.



Traditional, Noise2Noise and Noise2Void training schemes side by side, with N2V single-image blind-spot result on the right. [N2V19 · Fig. 1]

ASPECT	NOISE2NOISE (2018)	NOISE2VOID (2019)	WHY IT MATTERS
Requires	Two noisy shots per scene	One noisy image. That is all.	Applicable to any archive, any moving subject, any single acquisition
Key innovation	Corrupted target with matching mean	Blind-spot receptive field	Prediction for pixel i uses its neighbourhood except pixel i
Why it cannot cheat	Target is independent of input	Noise is conditionally pixel-wise independent, so neighbours carry no information about n_i	The identity solution becomes unreachable by construction

N2V19 §3.1, §3.4, §3.5, §4.1, Fig. 4 · N2N18 Table 1



N2V vs N2N — removing the second photograph CONTINUED

ASPECT	NOISE2NOISE (2018)	NOISE2VOID (2019)	WHY IT MATTERS
Implementation change	Swap the target tensor	Replace the centre pixel with a random neighbour; score the loss only there	A masking trick, not an architecture — any standard CNN acquires the property
Performance (BSD68, $\sigma=25$)	28.86 dB	27.71 dB	−1.15 dB vs N2N, and 0.88 dB below BM3D (28.59 in the same table) — the cost is real and the paper does not hide it
Applications	Burst photography, rendering	Cryo-TEM, fluorescence microscopy, cell tracking	Domains where the paper own figure captions the ground-truth column does not exist

THE HONEST FRAMING — AND THE ONE TO USE IF CHALLENGED

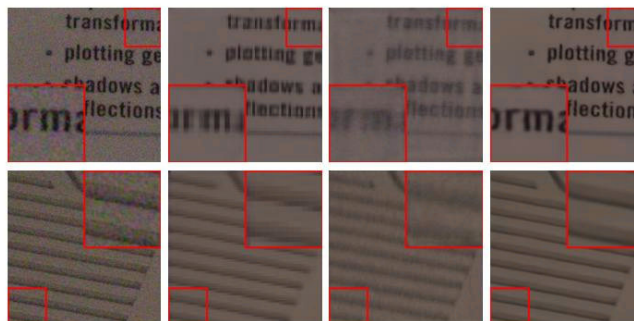
Self-supervision is not a quality technique, it is an availability technique. Where clean data exists, supervision wins by **1.35 dB** — the supervised baseline scores 29.06 against N2V 27.71, both numbers from N2V own comparison [N2V19]. Where it does not, the comparison is not 27.71 against 29.06 — it is 27.71 against nothing at all.

BM3D is 28.57 in DnCNN17 Table II and 28.59 in N2V19 comparison — we quote each table own number and never mix the two.



Down to one image, then down to no sacrifice

The branch finishes by removing the last two requirements. Compressed into one slide because the pattern is now familiar — each paper removes exactly one thing.



ZS-N2N [27] MASH [11] ATBSN [8] Ours

The information-lossy barrier made visible: ZS-N2N, MASH and ATBSN all lose fine structure that P2N renoise-and-agree supervision keeps. [P2N25 • Fig. 2]

ZS-N2N (2023)

Requirement removed — The training set. No corpus, no pretraining.

Key innovation — A pair downsampler: two fixed diagonal filters split one image into a noisy pair with similar signal and independent noise.

Implementation change — Two fixed stride-2 convolutions; a 20,000-parameter two-layer net fitted to that one image; residual + consistency loss.

P2N (2025)

Requirement removed — The information loss itself.

Key innovation — RDC: renoise the model own output with + and −, giving y_+ and y_- at full resolution. DCS: train the two denoised results to agree.

Implementation change — Three forward passes per step, on tensors that do not exist until the model has run once. Output compared against output, not against a target.

Down to one image, then down to no sacrifice CONTINUED

ZS-N2N (2023)

Why it is not just a new loss — It is a new input construction.

Cost paid — Half the spatial resolution during training; 1–2 K gradient steps per image.

Applications — Single archival images, one-off scientific captures; ~1 min on CPU vs Self2Self 1.2 h on GPU.

P2N (2025)

Why it is not just a new loss — Same: a loss scores a fixed prediction against a fixed target. P2N changes what goes in, how many passes, and which pair is compared.

Cost paid — Three passes per step; still per-image test-time optimisation.

Applications — Self-supervised denoising where masking artefacts are unacceptable.

The through-line: the network never moved, the target did.

Clean image (2018)



second copy (2019)



training set (2023)



information loss (2025)

Four papers, four requirements removed, one U-Net throughout.

4

PART 4 / 5

| BRANCH B • 2021 - 2024

Replacing the Convolution

- 01 The evidence for bottleneck 3
- 02 Restormer — attention on a different axis
- 03 NAFNet — delete the mechanisms
- 04 SCUNet and MambaIR
- 05 Why there are several SOTAs

The other debt: a 35×35 receptive field, and three competing operators that fix it.

brief §4

2007

2014

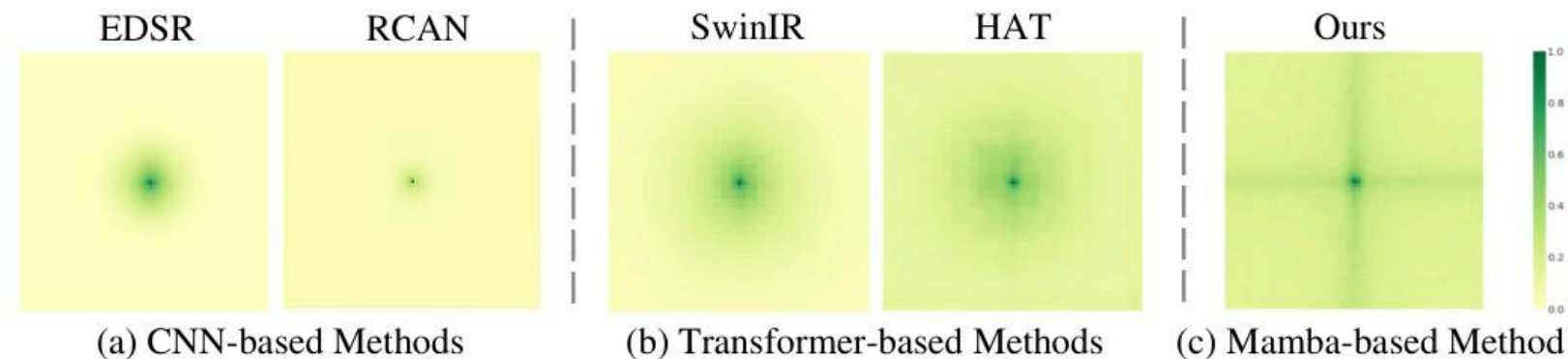
2017

2022

2025

One number points to the receptive field as the problem

Identical models, identical table, identical noise level. The only thing that changes is the dataset — and the gain changes by a factor of five.



Effective receptive field: CNNs stay tightly local, Transformers spread further, the Mamba-based backbone activates the widest field. [MambaIR24 · Fig. 1]

+0.39 dB

BSD68 · $\Sigma=50$ · DNCNN → RESTORMER

26.23 → 26.62, blind models, one table. Ordinary natural images, little long-range repetition: five years of architecture research buys almost nothing.

+1.98 dB

URBAN100 · $\Sigma=50$ · SAME TWO MODELS

26.35 → 28.33, same table. Photographs of buildings — windows and railings repeating across hundreds of pixels. Five times the gain, on the dataset made of repetition.

-0.71 dB

BARBARA · $\Sigma=25$ · DNCNN VS BM3D · 2017

The same phenomenon five years earlier, in DnCNN own results — the one Set12 image dominated by repetitive texture.

Restormer22 Table 4 (the four $\sigma=50$ numbers, single table) · DnCNN17 Table III

2007

2014

2017

2022

2025

One number points to the receptive field as the problem CONTINUED

- 01 **What this suggests.** The architecture branch did not make denoising uniformly better. It made denoising better specifically where non-locality matters — which is exactly the capability a 3×3 convolution stack structurally lacks.
- 02 **What it does not prove.** The comparison changes several variables at once: capacity, normalisation, and five years of training practice. NAFNet own ablation (slide 32) will show LayerNorm alone is worth +0.44 dB. The honest claim: the gain pattern is consistent with the receptive-field hypothesis — evidence, not proof.
- 03 **Why depth cannot fix it.** Receptive field grows linearly with depth: $(2d+1)^2$. Reaching WNNM 361×361 the DnCNN way needs about 180 layers. The fix had to be a different operator, not more of the same one.

Provenance: the 0.39-vs-1.98 comparison is our own analysis of published values, both pairs taken from Restormer Table 4 so the protocol is identical. Say computed from Table 4 if challenged.

Attention becomes affordable by changing one axis

Zamir et al., CVPR 2022. Standard self-attention costs $O(W^2H^2)$ — a 256×256 image needs a 4.3-billion-entry attention matrix, against 196 tokens for a ViT. Restormer attends over channels instead of pixels.

Restormer/basicsr/.../restormer_arch.py — Attention.forward

```
# BEFORE — spatial attention (ViT / SwinIR)
'b c h w -> b head (h w) c' → q@kT is (HW × HW)
256×256 → 65,536 × 65,536

# AFTER — Restormer MDTA
q = rearrange(q, 'b (head c) h w -> b head c (h w)',
              head=self.num_heads) # ← pixels on the LAST axis
k = rearrange(k, ...); v = rearrange(v, ...)
q = F.normalize(q, dim=-1); k = F.normalize(k, dim=-1)
attn = (q @ k.transpose(-2,-1)) * self.temperature
# → (C/head × C/head). At level 1: 48 × 48, any image size
```

- 01 **One einops pattern is the entire complexity claim.** Putting pixels on the last axis makes `q @ k.transpose(-2, -1)` a $C \times C$ matrix — 48×48 whether the image is 256² or 4000×3000. Cost becomes linear in pixels.
- 02 **Is it really global? Honestly: not pairwise-pixel.** Every channel is computed from every pixel, so channel statistics carry image-wide information; the paper says global relationships are implicitly modelled and never claims otherwise.
- 03 **The depth-wise convolutions are load-bearing, not decoration.** Removing `qkv_dwconv` costs 0.15 dB on Urban100 $\sigma=50$ — more than most architecture papers gain in total. [Restormer22 Table 7]
- 04 **Why SOTA:** the first method past 40 dB on both real-noise benchmarks — SIDD 40.02 / DND 40.03, the only one to do so in its own comparison table [Restormer22 Table 6]. Leads Urban100 at every noise level [Table 4]; 3.14× fewer FLOPs and 13× faster than SwinIR. Published at CVPR 2022 — how long it held those tables is not something any source states.

measured — swz30/Restormer, verbatim · Restormer22 §1, §3.1, Tables 4, 6, 7

2007

2014

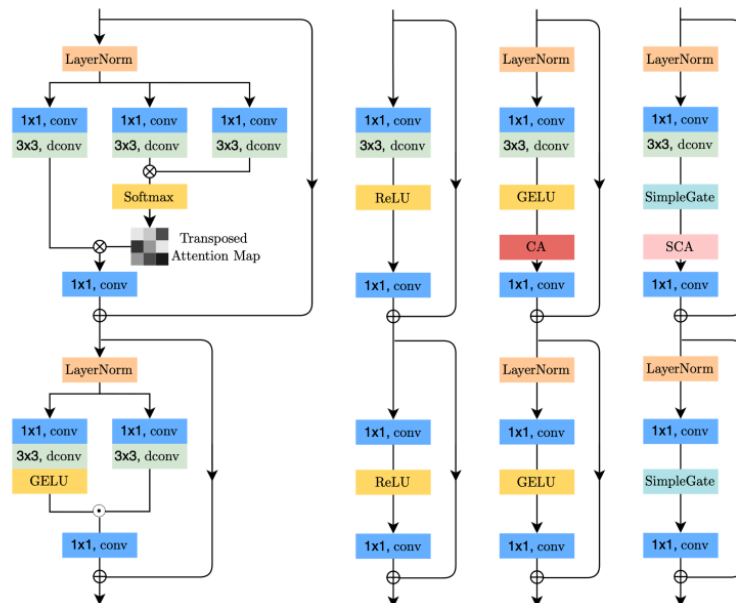
2017

2022

2025

The paper that asked whether any of it was necessary

Chen et al., ECCV 2022. While everyone added mechanisms, NAFNet removed them and scored higher. Its derivation is the most elegant reasoning in the modern half of this story.



The lineage NAFNet derives in one figure: the transformer block simplified step by step down to SimpleGate and SCA. [NAFNet22 · Fig. 3]

The method: take a standard transformer restoration block and delete every component you cannot justify — the activation, the softmax, the sigmoid. Each deletion is argued from first principles, and each deletion improves the score. [NAFNet22 §4]

The paper that asked whether any of it was necessary CONTINUED

```
NAFNet/basicsr/.../NAFNet_arch.py
```

```
# BEFORE — GELU, then a gate
```

```
GELU(x) = 0.5x(1+tanh[sqrt(2/pi)(x+0.044715x³)])
```

```
CA(X) = X · sigmoid(W2 relu(W1 pool(X)))
```

```
# AFTER — SimpleGate: the gate IS the nonlinearity
```

```
def forward(self, x):
```

```
    x1, x2 = x.chunk(2, dim=1)
```

```
    return x1 * x2 # ← replaces GELU entirely
```

```
# AFTER — SCA: no sigmoid, no relu
```

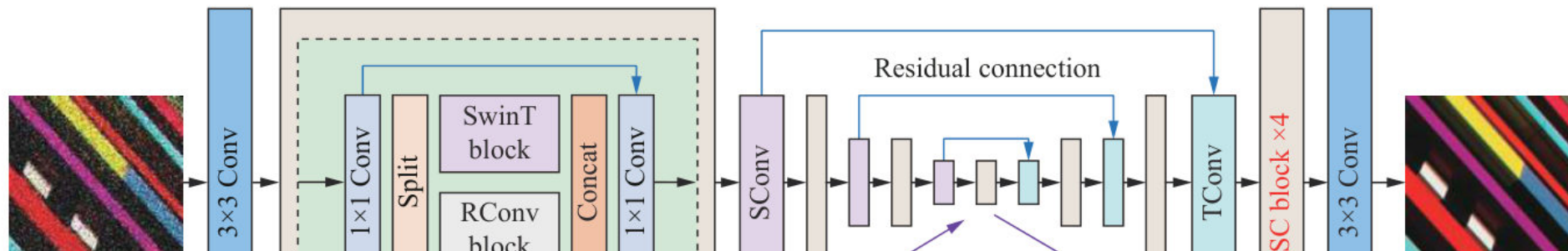
```
self.sca = nn.Sequential(nn.AdaptiveAvgPool2d(1),  
                        nn.Conv2d(c, c, 1))
```

- 01 **The derivation.** GELU is a special case of a gated linear unit. But a GLU itself contains nonlinearity: $f(X) \odot g(X)$ is nonlinear because a product of two linear maps is not linear. So delete the activation — the gate already supplies it.
- 02 **Apply the same argument to channel attention** and both its nonlinearities vanish too. Result: no Sigmoid, no ReLU, no GELU, no Softmax anywhere — the first work demonstrating that nonlinear activation functions may not be necessary for SOTA computer vision methods. [NAFNet22 abstract]
- 03 **Both simplifications improved the score.** SimpleGate beats GELU (+0.08 dB), SCA beats CA (+0.03 dB) on SIDD. [NAFNet22 §4]
- 04 **The uncomfortable finding in their own ablation:** the largest single gain is LayerNorm, +0.44 dB (39.29 → 39.73) — because it stabilises training at a 10× larger learning rate. A large share of reported architectural progress in restoration is optimisation progress wearing a costume. [NAFNet22 §4]

Why SOTA: 40.30 dB on SIDD — the highest real-noise result in our comparison, +0.28 dB over Restormer 40.02 — at less than half the previous best compute. Published at ECCV 2022. And by far the simplest code to maintain. [NAFNet22 abstract]

Two more answers — one architectural, one not about architecture at all

The branch does not end with Transformers. MambaIR proposes a different operator again; SCUNet argues the operator was never the problem.



The Swin-Conv-UNet pipeline: swin-conv blocks combine local and non-local modelling on the way down and back up a UNet backbone. [SCUNet23 · Fig. 1]

MAMBAIR (2024)

The claim — Attention is quadratic; windowing surrenders globality. Borrow a selective state-space model — **global receptive field at linear cost**.

Technical change — S6 recurrence with input-dependent B, C, Δ . Dual form: recurrent (unbounded memory) for reach, convolutional for parallel training.

SCUNET (2023)

The claim — The benchmark is the problem. A model trained on AWGN performs poorly for most of real images due to **noise assumption mismatch**.

Technical change — Keep local and non-local priors in parallel: split the feature map, half to a Swin block, half to a residual conv block.

Two more answers — one architectural, one not about architecture at all CONTINUED

MAMBAIR (2024)

Implementation change — Flatten the image to a sequence; four-directional scan; add a local conv (fixes local pixel forgetting) and channel attention (fixes redundancy).

Why SOTA / where — Best DND result, 40.04 dB — by 0.01 dB over Restormer 40.03 [MambaIR24 Table 6]. Leads super-resolution on Manga109 $\times 2$ by 0.41 dB over SwinIR.

Cost / limitation — Will not import without compiled CUDA extensions. Its own changelog records that published thop complexity figures were wrong; the denoising model parameter count is not reported in the paper. [measured]

SCUNET (2023)

Implementation change — The dataloader becomes the contribution: Gaussian, Poisson, speckle, JPEG and sensor noise synthesised through a real inverted ISP, plus resizing and double degradation.

Why SOTA / where — Trained purely on synthetic mixed degradation (no SIDD/DND pairs). Among blind models on standard colour denoising it edges Restormer where non-locality matters: Urban100 $\sigma=50$ 30.14 vs 30.02, CBSD68 $\sigma=50$ 28.61 vs 28.60 [SCUNet23 comparison table]. Not top of any AWGN leaderboard — and it does not need to be.

Cost / limitation — 9.66 M parameters, smallest of the four, built to survive real photographs. Its abstract still concedes a general-purpose blind real denoiser remains unsolved.

The prior is a file — the whole pipeline in practice

What the brief calls the practical half: the data pipeline, the loss functions that create the file, and the three lines that load it back.

THE LOSS THAT CREATES THE FILE — one row per era

ERA	DATA	LOSS
Supervised — DnCNN, Restormer, VAINet	pairs (y1, y2): synthetic AWGN or real SIDD pairs	$\ R(y) - (y-x)\ ^2$ or L1
Noise2Noise	pairs (y1, y2): two noisy shots	same MSE — targets swapped
Noise2Void	single noisy y	MSE scored only at masked centre pixels
ZS-N2N	one image → two downsampled views	residual + consistency loss
P2N	model output renoised ±	agreement loss — no target at all

loading the pretrained prior

```
# 1 — the prior is a 2.7 MB file (0.67 M params, fp32)
net = DnCNN(in_nc=1, out_nc=1, nc=64, nb=17, act_mode=R)
ckpt = torch.load('dncnn_sigma25.pth', map_location='cpu')
net.load_state_dict(ckpt)
net.eval() # freeze batch-norm statistics

# 2 — one noisy image through the trained prior
y = torch.from_numpy(img).permute(2,0,1).unsqueeze(0)/255.
with torch.no_grad(): # no optimisation at test time
    n = net(y) # R(y): the noise, learned
    out = (y - n).clamp(0,1) # x = y - R(y) → return x-n

# 3 — Hugging Face variant, same pattern:
# from huggingface_hub import snapshot_download
# path = snapshot_download(repo_id) # → local .pth
```

Every era of this talk in one line: **classical** = solve an optimisation per image; **deep** = load_state_dict once, forward in 60 ms; **self-supervised** = the same file, fitted to one image.

Why There Is No Single Winner

- 01 The SOTA comparison table
- 02 The same crop through every era
- 03 Why several coexist
- 04 Applications and trade-offs
- 05 Open problems and proposals

The question this talk was built to answer, now with the numbers behind it.



Six methods — each leads a different table

Every row names the benchmark and the metric — a claim without those is not checkable. Quality figures reproduced from published tables; parameters and CPU latency are our own measurements.

METHOD	LEADS ON	HEADLINE FIGURE	PUBLISHED	WHAT DISTINGUISHES IT	COST
NAFNet (2022)	SIDD — real sensor noise	40.30 dB	ECCV 2022	No attention, no activation functions at all; simplest code	116.0 M · 1.90 s CPU
Restormer (2022)	Urban100 — long-range structure	$\sigma=50$: 28.33 SIDD 40.02 / DND 40.03	CVPR 2022	First past 40 dB on both real-noise benchmarks; channel attention at linear cost	26.1 M · 10.44 s CPU
MambaIR (2024)	DND — held-out real noise	40.04 dB (+0.01)	ECCV 2024	Global receptive field at linear cost via selective state-space scan	CUDA-compiled ext. to import; DN params not in paper
SCUNet (2023)	Its own blind mixed-noise benchmark — no standard table	Urban100 $\sigma=50$: 30.14 top among blind models	Mach. Intell. Res. 2023	Competes on the training distribution, not the architecture	9.7 M · 1.69 s CPU
DnCNN (2017)	Historical — BSD68 Gaussian	29.23 dB ($\sigma=25$)	IEEE TIP 2017	The reference the whole field was built against	0.67 M · 1.02 s CPU
BM3D (2007)	Zero-shot — no training data	28.57 dB ($\sigma=25$)	IEEE TIP 2007	No training, no GPU; still the zero-shot baseline in N2N18, N2V19, ZS-N2N23	No training · 2.85 s

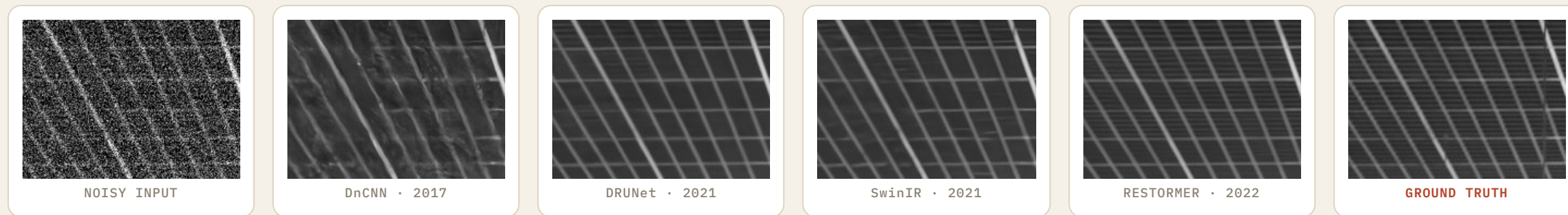
No source states a SOTA window, so this deck does not assert one. Read the Leads-on column: these methods are not ranked on one axis — they lead different benchmarks, and those benchmarks measure different capabilities.



The same crop through every era

Quantitative tables rank; crops reveal. Two published comparisons, one image each — every tile reproduced verbatim from the cited figure.

GAUSSIAN DENOISING — ONE CROP, FOUR GENERATIONS • [RESTORMER22 • FIG. 6, TOP ROW]



REAL SENSOR NOISE — SIDD • [NAFNET22 • FIG. 5]



- 01 **The top row is the architecture branch made visible:** as the receptive field widens (slide 28), the repeating texture returns — DnCNN leaves it smooth, DRUNet and SwinIR recover part of it, Restormer restores it.
- 02 **The bottom row is the real-noise story:** 0.8 dB separate MPRNet from NAFNet in PSNR, but the texture difference is what a practitioner sees first — the case slide 5 made for reporting SSIM and LPIPS alongside PSNR.
- 03 **Method note:** crops are reproduced verbatim from the published figures cited below — same image per row, no cherry-picking on our side.

Restormer22 Fig. 6 (top row) • NAFNet22 Fig. 5 — tiles reproduced from the published figures

2007

2014

2017

2022

2025

Four reasons the field has not converged on one method

This is the question the professor is most likely to ask, so it gets its own slide rather than a sentence.

REASON 1

The benchmarks measure different things

BSD68 measures average-case Gaussian removal and has saturated: $+0.35$ dB across five years at $\sigma=25$ (DnCNN-B $29.16 \rightarrow$ Restormer 29.51 , both blind models, one table [Restormer22 Table 4]). Urban100 measures long-range self-similarity; SIDD and DND measure real sensor noise. A method can lead one and trail another without any contradiction.

REASON 2

The cost axes disagree with each other

Our own measurements give three incompatible rankings of the same models: by parameters DnCNN wins; by CPU wall-clock the P2N backbone wins; by real-noise quality NAFNet wins. NAFNet has $4.4\times$ Restormer parameters and runs $5.5\times$ faster. More efficient is meaningless without naming the metric and the device.

REASON 3

Deployment constraints are not quality constraints

MambaIR leads DND by 0.01 dB and cannot be imported without compiled CUDA extensions. SCUNet leads no standard leaderboard and is a third of Restormer size. Which one is best depends entirely on what you are shipping onto what hardware.

REASON 4

Supervision availability partitions the field

Branch A and branch B answer different questions. Where clean pairs exist, supervision wins by 1.35 dB — 29.06 vs 27.71 , both numbers from N2V own comparison [N2V19]. Where they do not, the self-supervised methods are not worse — they are the only option, and the supervised SOTA is simply unavailable.

What you would actually deploy, and why

The honest answer to which is best names the constraint first. Five constraints, five different answers, all current — which is what an unresolved research question looks like when you have to ship something.

IF YOUR CONSTRAINT IS...	DEPLOY	BECAUSE	ACCEPT
Real photos, GPU, paired data	NAFNet	40.30 dB on SIDD; best accuracy-per-millisecond we measured; simplest code to maintain	116 M parameters
Repetitive structure, max quality	Restormer	Leads Urban100 at every σ [Table 4]; first method past 40 dB on both real-noise benchmarks (SIDD 40.02 / DND 40.03)	10.4 s per 256 ² on CPU
CPU or memory bound	SCUNet	9.7 M parameters; synthetic-degradation training survives resized and JPEG-compressed input	Not top of any AWGN leaderboard
No clean data exists at all	P2N / ZS-N2N	Microscopy, cryo-EM, astronomy — where ground truth cannot be acquired	Per-image test-time optimisation
No data, no GPU, no dependencies	BM3D	Seventeen years old and still beats AWGN-trained DnCNN on real noise by 1.99 dB [Restormer22 Table 6]	2.85 s, and a quality ceiling

Denoising is rarely the product: it is preprocessing for detection and segmentation — where the metric that matters is downstream IoU (slide 5) — and a plug-and-play prior inside other solvers. [SCUNet23 §1 · DnCNN17 §III-D]



What eighteen years did not solve

Each of these is grounded in a documented limitation from the papers themselves or in our own measurements — not speculation.

OPEN 1

A blind denoiser for real cameras

SCUNet abstract states it remains unsolved. Neither synthetic degradation nor SIDD-trained models generalise to an unseen camera with an unseen ISP, and no benchmark currently measures that. DnCNN scores 23.66 dB on SIDD [Restormer22 Table 6] — 5.6 dB below its own BSD68 score, and 16.4 dB behind Restormer 40.02 in the same table. Synthetic training does not survive contact with real sensor noise: a cliff, not a graceful degradation.

OPEN 2

Fidelity metrics reward blur

Every loss in the modern field is L1, L2 or PSNR, and all three recover a conditional mean — which is smoother than any plausible clean image. Noise2Noise names the mechanism itself: **L2 training results in spatial blurriness**. While leaderboards rank by PSNR, the incentive points the wrong way. [N2N18 §2]

OPEN 3

Nobody agrees what efficient means

Three incompatible rankings from our own measurements, and MambaIR changelog concedes its published complexity figures were wrong. Efficiency claims across papers are close to uncheckable.

OPEN 4

The benchmarks have saturated

At $\sigma=50$, BSD68 moved 0.39 dB in five years while Urban100 moved 1.98 — five times as much [Restormer22 Table 4]. On SIDD, the DnCNN-to-Restormer gap in one table is 16.4 dB [Restormer22 Table 6]. A benchmark that cannot distinguish 2017 from 2022 is no longer measuring what the field works on.

Four research directions the limitations actually justify

Each is tied to a specific documented gap, states what would change, and names how it would be measured. None of these claims a paper achieved something it did not.

DIRECTION	GROUNDING IN	WHAT WOULD CHANGE	HOW YOU WOULD MEASURE IT
Cross the fork: self-supervised supervision on a modern backbone	Vision on a modern backbone: our work trains a Transformer or SSM backbone under RDC/DCS supervision; P2N still ships the 2018 U-Net	Replace UNet_n2n_un with a Restormer or NAFNet block under P2N RDC/DCS training signal	N2V own protocol on BSD68 $\sigma=25$ and on cryo-TEM / CTC sets — does the 1.35 dB gap (slide 24) shrink?
Cross-ISP generalisation benchmark	SCUNet abstract: a general blind real denoiser remains unsolved, and no benchmark tests unseen cameras	Train on SIDD, evaluate on DND and a held-out camera; report the drop as the headline number	Δ PSNR between seen and unseen ISP — a metric nobody currently reports
Perceptual objective without losing fidelity	Noise names L2 spatial blurriness; perceptual losses are common in SR and absent from denoising	Add an LPIPS or no-reference term to NAFNet PSNR loss; report both axes, not one	PSNR and LPIPS jointly on SIDD — accept a small dB loss for a measured perceptual gain
An efficiency protocol the field can agree on	Agree on compatible rankings from our measurements; MambaIR corrected thop figures	Report parameters, MACs and wall-clock on a named device, with the counting method stated explicitly	Reproduce the table across CPU and GPU; the ratio changes are themselves the result

The first row is the one we would pursue. It falls directly out of the structural claim of this project — the two branches have never merged — and it is cheap to test, because both halves already exist as working code.

grounded in P2N25, SCUNet23 abstract, N2N18 §2, MambaIR24 changelog, and our own measurements



The evolution, in one paragraph

Denoising is the cleanest case study in how computer vision replaced hand-designed priors with learned ones, because here **the prior is written down explicitly and you can watch it move.**

Four eras, three bottlenecks. Manual prior design ended when DnCNN made the prior a downloadable file. That created two new debts — it needed clean data, and it could only see 35 pixels — and in 2018 the field forked to pay them off separately.

Neither branch has adopted the other advance. And because three architectures now lead three different benchmarks while none solves real-camera generalisation, **the story ends in an open question rather than a winner.**

— *What the evidence in this talk supports.*

the organising claim, carried since Stage 1 · evidence: DnCNN17 · N2N18 · N2V19 · Restormer22 · NAFNet22 · SCUNet23 · MambaIR24



Every equation this talk leaned on

Kept for the defence: the formulas named on the method slides, written as equations with their sources.

MAP — THE CLASSICAL PRIOR

$$\hat{x} = \operatorname{argmin}_x \frac{1}{2} \|y - x\|^2 + \lambda \cdot \Phi(x)$$

The prior written down: explicit, differentiable, arguable.

[DnCNN17 §I]

WNNM — WEIGHTED NUCLEAR NORM

$$\hat{X} = \operatorname{argmin}_X \frac{1}{2} \|Y - X\|_F^2 + \sum_i w_i \cdot \sigma_i(X), \\ w_i \propto 1/\sigma_i$$

Large singular values carry the image and are shrunk less.

[WNNM14 §3, Thm 2]

DnCNN — RESIDUAL LEARNING

$$\ell(\theta) = 1/2N \sum_i \|R(y_i; \theta) - (y_i - x_i)\|^2$$

The network learns v , not x ; the output is $y - R(y)$.

[DnCNN17 Eq. (1)]

NOISE2NOISE — THE THEOREM

$$\operatorname{argmin}_{\theta} E \|f(y; \theta) - z\|^2 \Rightarrow f^* = E[z|y]; \\ E\{z'\} = E\{z\} \Rightarrow \text{same optimum}$$

A zero-mean corrupted target is as good as a clean one.

[N2N18 Eq. (2)–(6)]

RESTORMER — MDTA

$$\text{Attn} = \operatorname{softmax}(\tau \cdot QK^T) \in \mathbb{R}^{C \times C}, \quad \text{out} = \\ V \cdot \text{Attn}^T$$

Pixels on the last axis: attention over channels, cost linear in HW.

[Restormer22 §3.1]

MAMBA — S6 RECURRENCE

$$h_t = \bar{A}h_{t-1} + \bar{B}_t \cdot x_t, \quad y_t = C_t \cdot h_t; \quad \bar{A} = \\ \exp(\Delta_t \cdot A), \quad \bar{B}_t = \Delta_t \cdot B_t$$

Selective state space: Δ, B, C are input-dependent — global reach at linear cost.

[MambaIR24 §2 · Gu & Dao 2023]

Where to find the answer to what you are about to be asked

Kept for the Q&A. The full evidence for every claim in this talk is in the 100-slide master deck, indexed here by the question likely to prompt it.

IF ASKED...	ANSWER IN ONE LINE	MASTER DECK
Why is 0.6 dB a big deal?	Seven years of classical refinement bought +0.11–0.39 dB over BM3D at $\sigma=25$; DnCNN took +0.66 dB in one paper [DnCNN17 Table II]	10, 37
Why predict noise, not the image?	y is closer to x than to v , so learning the image forces an identity mapping — the hard point in parameter space	33
Why do RL and BN need each other?	RL makes layer inputs Gaussian-like and content-independent — what BN batch statistics assume	34
Why depth 17?	$RF = (2d+1)^2$ set to 35×35 to match EPLL, the smallest effective patch size in the field	35
How can noisy targets work?	L2 learns the conditional mean; the mean of a zero-mean corrupted target is the clean target	43
Where is Restormer attention linear?	<code>Attention.forward</code> — the rearrange puts pixels last, so $q@kT$ is $C \times C$ not $HW \times HW$	69
Why is NAFNet faster with more parameters?	1×1 and depth-wise convs hit optimised GEMM paths; rearrange forces tensor-layout changes	77
What did the code tell you that the papers did not?	Scaling constants differ $8\times$; SPI cuts parameters and triples latency; MambaIR needs CUDA to import	75, 77
Which would you deploy?	Name the constraint first — five constraints, five answers, all current	89
Did you verify the PSNRs yourself?	No. Every quality figure is reproduced from a named published table; only parameters and latency are ours	81, 93
Where are the equations?	Slide 44 of this deck — WNNM, MDTA, S6 and the rest in one place	this deck, 44

The rule for the defence: do not put anything on a slide you cannot personally explain. Every claim in this talk traces to a named table in a named paper, or to a measurement whose method is stated.

Q&A index · master-deck numbers refer to the 100-slide deck submitted separately

2007

2014

2017

2022

2025